

A COMPARISON OF APPROXIMATE CONFIDENCE INTERVAL METHODS FOR THE DIFFERENCE BETWEEN TWO INDEPENDENT BINOMIAL PROPORTIONS

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Summary: Conventional methods to determine confidence intervals for the difference between two independent binomial proportions p_1 and p_2 are prone to violations of the definition interval $[-1; 1]$ for $p_1 - p_2$ and may have very poor coverage properties. In this paper several less known methods are described. A simulation study was done to compare the different confidence interval methods with respect to length, coverage, zero width interval and violation of the definition interval. The best methods were found to be Mee's and Miettinen and Nurminen's method. These methods, however, are

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computer intensive. The Jeffreys-Perks and Score interval methods seem to be the best of the more easily calculable methods to use in most practical situations.

1. Introduction

The comparison of proportions in two independent groups is often encountered in medical data. Confidence intervals for the difference $b = p_1 - p_2$ (also called the risk difference), the odds ratio $\psi = p_1(1 - p_2) / (1 - p_1)p_2$ and the relative risk $\rho = p_1/p_2$ of two Binomial proportions are provided by most statistical software. Unfortunately, only asymptotic confidence intervals for the difference of two Binomial proportions are available in these software packages. Similarly, standard textbooks usually only provide the asymptotic intervals. Brenner and Quan (1990) noted that it has long been known that the asymptotic interval methods are problematic. In general asymptotic intervals are satisfactory only for large samples and proportions close to 0.5 (Altman, 1992, Rosner, 1990 and Zar, 1984).

Problems that can occur when using the asymptotic interval method are (i) intervals of zero length, (ii) violation of the definition interval $[-1; 1]$ and (iii) unsatisfactory coverage.

The purpose of this paper is to compare the coverage and length of the confidence intervals calculated by using the well known and less well known confidence interval methods for the difference between two independent Binomial proportions. Newcombe (1998) also did a simulation study on different confidence interval methods. Since this article two papers have appeared with new methods, namely Wallenstein (1997) and Feigin and

Lumelskii (2000). Our simulation and comparisons take these new methods into accounts. Recommendations are made as to which interval methods should be used for different sample sizes. Suitable interval methods are recommended with regard to the ease and speed of computation.

2. Methods to calculate confidence intervals for the difference between two proportions

The following methods for calculating confidence intervals for the difference $b = p_1 - p_2$ between two proportions p_1 and p_2 from two independent Bernoulli distributions with probability functions $f_i(z) = p_i^z (1 - p_i)^{1-z}$ where $z = 0, 1$ and $i = 1, 2$ will be considered, based on random samples of sizes n_1 and n_2 . If X_i denotes the number of successes $X_i = \sum_{j=1}^{n_i} Z_{ij}$, then X_i has a binomial distribution, which we will denote as $X_i \sim \text{Bin}(n_i, p_i)$.

2.1 Asymptotic approximate confidence interval

The usual asymptotic approximate $100(1 - \alpha)\%$ confidence interval for the difference between the proportions p_1 and p_2 is denoted as

$$\text{CONF}(p_1 - p_2)_{1-\alpha} = (\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}} = [l_1; l_2],$$

where l_1 results by using the minus sign and l_2 by using the plus sign and where $\hat{p}_i = \frac{x_i}{n_i}$, x_i = number of successes from population i , $i = 1, 2$,

n_i = number of repetitions from population i , $i = 1, 2$,

and $z_{\alpha/2}$ = relevant $(1 - \alpha/2)$ 100 percentile of the normal distribution.

This "conventional confidence interval" is easy to calculate and is used in the software SAS® Procedure FREQ (SAS Institute Inc. 1997) as the method to calculate a confidence interval for the difference $p_1 - p_2$.

2.2 Asymptotic approximate confidence interval with Yates's continuity correction

$$CONF (p_1 - p_2)_{1-\alpha} = (\hat{p}_1 - \hat{p}_2) \pm \left[z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2} + \left(\frac{1}{2n_1} + \frac{1}{2n_2} \right)} \right].$$

(Yates, 1934)

2.3 Hauck and Anderson interval

Hauck and Anderson (1986) proposed another modification of the conventional asymptotic approximate confidence interval given in 2.1.

$$CONF (p_1 - p_2)_{1-\alpha} = (\hat{p}_1 - \hat{p}_2) \pm \left[z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1-1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2-1} + \frac{1}{2\min(n_1, n_2)}} \right].$$

A unified method to improve the conventional approximate confidence interval for $p_1 - p_2$ has been proposed by Beal (1987). If $X_1 \sim \text{Bin}(n_1, p_1)$ independent of $X_2 \sim \text{Bin}(n_2, p_2)$, the following reparameterization is useful: $a = p_1 + p_2$; $b = p_1 - p_2$; $u = (1/4)(1/n_1 + 1/n_2)$; and $v = (1/4)(1/n_1 - 1/n_2)$. Let $\hat{a}$, $\hat{b}$, $\hat{p}_1$ and $\hat{p}_2$ denote the maximum likelihood estimators of a , b , p_1 and p_2 and $V(a, b; u, v) = \text{Var}(\hat{b}) = u[(2-a)a - b^2] + 2v(1-a)b$.

The approach to find confidence intervals for $b = p_1 - p_2$ is based upon solving for b from the equation

$$(b - \hat{b})^2 = c \text{Var}(\hat{b}) = cV(\hat{a}, \hat{b}; u, v), \quad (2.1)$$

where $c = \chi_{1, \alpha}^2$ and $\chi_{f, \alpha}^2$ denotes the relevant $100(1 - \alpha)$ percentile of the Chi-square distribution with f degrees of freedom and where $\hat{a}$ and $\hat{b}$ are expressions for a and b which do not necessarily contain the variable b . Different confidence intervals can now be calculated by solving for b in equation (2.1) by choosing different values for $\hat{a}$ and $\hat{b}$ (see methods 2.4, 2.5, 2.6 and 2.7 below).

2.4 Jeffreys-Perks interval

Using a Bayesian approach, this interval is calculated as follows (Beal, 1987):

$$CONF(p_1 - p_2)_{1-\alpha} = \hat{b} - \frac{\hat{b} + cu(1-\hat{a})}{1+cu} \pm \left[z_{\alpha/2} \frac{\sqrt{V(\hat{a}, \hat{b}; u, v) + cu^2(2-\hat{a})\hat{a} + cv^2(1-\hat{a})^2}}{1+cu} \right],$$

where $c = z_{\alpha/2}^2$ and $\hat{a} = 1/2 \left(\frac{x_1 + 1/2}{n_1 + 1} + \frac{x_2 + 1/2}{n_2 + 1} \right)$.

2.5 Haldane interval

Beal (1987) mentions an interval similar to the Jeffreys-Perks interval called the Haldane interval, but with $\hat{a} = \frac{x_1}{n_1} + \frac{x_2}{n_2}$.

2.6 Mee interval

Mee (1984) used a Bayesian approach to the estimation of $\hat{a}$ and $\hat{b}$ and noted that equation (2.1) can be used with $\tilde{a} = 2\hat{p}_2 + b$, $\tilde{b} = b$ and that the results obtained from the unified method and Mee's method differ. Mee suggested that equation (2.1) be solved with $\hat{b}$ chosen as $\hat{b} = b$ and $\hat{a} = a^*(b)$ or $\hat{a} = a^*$, the maximum likelihood estimator of $a = p_1 + p_2$ when $b = p_1 - p_2$ has a given value. Since $\hat{a} = a^*$ is now a function of b , equation (2.1) with these values is not quadratic and cannot be solved in closed form. Instead, equation (2.1) has to be solved numerically to obtain the two roots b_l and b_u . Since a^* is defined only in the interval $[0; 2]$ we seek the smallest value $b_l \in [-1; 1]$ and the largest value $b_u \in [-1; 1]$ such that $(b - \hat{b})^2 \leq cV(a^*, \hat{b}; u, v)$ for $b_l \leq b \leq b_u$. Then $L = b_l$ and $U = b_u$.

An iterative method can be used to calculate the upper and lower limits of the Mee interval as described by Beal (1987).

2.7 Miettinen and Nurminen interval

Miettinen and Nurminen (1985) used a similar approach to Mee except that the constant c in equation (2.1) is chosen as $cN/(N-1)$ so that equation (2.1)

becomes

$$(b - \hat{b}) = cN / (N - 1) \text{Var}(\hat{a}, \hat{b}; u, v),$$

where $N = n_1 + n_2$.

Like the Mee interval method this method requires intensive computation.

2.8 Wallenstein interval without continuity correction

Wallenstein (1997) mentioned that the conventional approximate interval is often too short and suggested a method to produce wider confidence intervals. He proposed a slight modification of Mee's procedure by using least squares estimates (LSEs) instead of MLEs for $\hat{p}_{1|d}$ and $\hat{p}_{2|d}$, and allowed for a continuity correction $\varepsilon \geq 0$. Here $\hat{p}_{i|d}$ is the estimate of p_i based on $b = d$ where d is the solution of the implicit equation

$$d = \hat{p}_1 - \hat{p}_2 \pm \left[z_{\alpha/2} \sqrt{\hat{V}(p_1, p_2)} \right] = [L; U]$$

where $\hat{V}(p_1, p_2) = \hat{p}_1(1 - \hat{p}_1)/n_1 + \hat{p}_2(1 - \hat{p}_2)/n_2$. Wallenstein proposed replacing $\hat{p}_1 - \hat{p}_2$ by $\hat{p}_1 - \hat{p}_2 + \varepsilon$ to obtain

$$d = (\hat{p}_1 - \hat{p}_2 + \varepsilon) + z_{\alpha/2} \sqrt{V(\hat{p}_{1|d}, \hat{p}_{2|d})} \quad (2.8.1)$$

The LSEs of p_i subject to the constraint $\hat{p}_{1|d} - \hat{p}_{2|d} = d$ are

$$\hat{p}_{1|d} = \tilde{p} + d \frac{n_2}{N}$$

and

$$\hat{p}_{2|d} = \tilde{p} - d \frac{n_1}{N}$$

where $\tilde{p} = (n_1 \hat{p}_1 + n_2 \hat{p}_2) / N$ and $N = n_1 + n_2$. For this interval the continuity correction is taken as $\varepsilon = 0$.

2.9 Wallenstein interval with continuity correction

In equation (2.8.1) ε is replaced by $\varepsilon = 1/2n_1 + 1/2n_2$. Wallenstein (1997) gave a detailed description on how to calculate the confidence intervals with and without a continuity correction.

2.10 Score interval without continuity correction

The following score intervals for $p_1 - p_2$ are based on Wilson's (1927) score interval for the single proportion, and were proposed by Newcombe (1998). This method combines Wilson's score intervals for each single proportion in much the same way as the conventional approximate method combines approximate intervals for the individual proportions. To obtain the Wilson score interval, first calculate

$$\hat{b} = \hat{p}_1 - \hat{p}_2 \quad \text{and then}$$

$$L_i = \left(\left(2n_i \hat{p}_i + z_{\alpha/2}^2 \right) - z_{\alpha/2} \sqrt{z_{\alpha/2}^2 + 4n_i \hat{p}_i \hat{q}_i} \right) / 2 \left(n_i + z_{\alpha/2}^2 \right),$$

$$\text{and } U_i = \left(\left(2n_i \hat{p}_i + z_{\alpha/2}^2 \right) + z_{\alpha/2} \sqrt{z_{\alpha/2}^2 + 4n_i \hat{p}_i \hat{q}_i} \right) / 2 \left(n_i + z_{\alpha/2}^2 \right)$$

for $i = 1, 2$.

$$\text{Substitute } L_i \text{ and } U_i \text{ in } \varepsilon_1 = z_{\alpha/2} \sqrt{\frac{L_1(1-L_1)}{n_1} + \frac{U_2(1-U_2)}{n_2}}$$

$$\text{and } \varepsilon_2 = z_{\alpha/2} \sqrt{\frac{U_1(1-U_1)}{n_1} + \frac{L_2(1-L_2)}{n_2}}.$$

Now let $L = \hat{b} - \varepsilon_1$ and $U = \hat{b} + \varepsilon_2$, then the $100(1 - \alpha)\%$ confidence interval for $p_1 - p_2$ is given by

$$CONF(p_1 - p_2)_{1-\alpha} = [L; U].$$

2.11 Score interval with continuity correction

The score interval can be improved with a continuity correction (Newcombe, 1998) as follows:

Firstly calculate $\hat{b} = \hat{p}_1 - \hat{p}_2$, and then

$$L_i = \frac{2n_i \hat{p}_i + z_{\alpha/2}^2 - 1 - z_{\alpha/2} \sqrt{z_{\alpha/2}^2 - 2 - \frac{1}{n_i} + 4\hat{p}_i n_i \hat{q}_i + 1}}{2 \left(n_i + z_{\alpha/2}^2 \right)}$$

and

$$U_i = \frac{2n_i\hat{p}_i + z_{\alpha/2}^2 + 1 + z_{\alpha/2}\sqrt{z_{\alpha/2}^2 + 2 - \frac{1}{n_i} + 4\hat{p}_i n_i \hat{q}_i - 1}}{2(n_i + z_{\alpha/2}^2)}$$

for $i = 1, 2$.

Substitute these values into

$$\epsilon_1 = \sqrt{\left(\frac{x_1}{n_1} - L_1\right)^2 + \left(U_2 - \frac{x_2}{n_2}\right)^2}$$

$$\text{and } \epsilon_2 = \sqrt{\left(U_1 - \frac{x_1}{n_1}\right)^2 + \left(\frac{x_2}{n_2} - L_2\right)^2}.$$

Let $L = \hat{b} - \epsilon_1$ and $U = \hat{b} + \epsilon_2$ which yield the confidence interval

$$CONF(p_1 - p_2)_{1-\alpha} = [L; U].$$

Note that if $x_1 = 0$ then $L_1 = 0$, if $n_1 - x_1 = 0$ then $U_1 = 0$. If $x_2 = 0$ then $L_2 = 0$, if $n_2 - x_2 = 0$ then $U_2 = 0$.

2.12 Belyaev and Lumelski

Feigin and Lumelskii (2000) suggested this method as well as the next method for calculating approximate confidence limits for the difference. The approximate interval in accordance with Belyaev (1966) and Lumelskii (1969) is as follows:

$$CONF(p_1 - p_2)_{1-\alpha} =$$

$$(\hat{p}_1 - \hat{p}_2) \pm \left[z_{\alpha/2} \sqrt{\frac{z_{\alpha/2}^2 + n_1 + n_2}{4n_1 n_2} - \frac{(\hat{p}_1 - \hat{p}_2)^2}{n_1 + n_2}} / \left(1 + \frac{z_{\alpha/2}^2}{n_1 + n_2} \right) \right].$$

2.13 Kolmogorov

This method is based on the approach of Kolmogorov (1950) for the approximate confidence interval method.

$$CONF(p_1 - p_2)_{1-\alpha} =$$

$$(\hat{p}_1 - \hat{p}_2) \pm \left[z_{\alpha/2} \sqrt{\sum_{i=1}^2 \frac{x_i(n_i - x_i)}{n_i^2(n_i - 1)}} + 1/4 \sqrt{\sum_{i=1}^2 \frac{n_i - 1}{n_i^3(n_i - 1.5)}} \right].$$

3. Simulation study

3.1 Description of the simulation study

The various interval methods were compared by a simulation study using the following algorithm:

1. Specify the parameter combination of proportions p_1 and p_2 and sample sizes n_1 and n_2 to be investigated.
2. Simulate pairs of 50 000 samples x_1 and x_2 given the specified parameter combinations of p_1, p_2, n_1 and n_2 .
3. Calculate confidence intervals for $p_1 - p_2$ for each of the 50 000 simulated samples using the different interval methods.
4. Determine the interval length, coverage, violation of the definition interval and zero width intervals of the different interval methods for each of the 50 000 samples.

The average length was determined as the sum of the differences between the upper bound and the lower bound for each simulation divided by the total number of simulations, whereas the coverage, violation of the definition interval and the occurrence of zero width intervals are expressed as a percentage out of 50 000 simulations.

The simulation study was done for the following choices of the proportions (p_1, p_2) : (0.6, 0.4), (0.95, 0.85), (0.95, 0.15), (0.98, 0.98) and (0.98, 0.02). The purpose of the choice of proportions is to cover cases for which the asymptotic methods are known to be unsatisfactory. The following sample

sizes were used: Sample sizes $n_1 = 100$ and $n_2 = 90$ were viewed as large. Sample size combination $n_1 = 60$ and $n_2 = 50$ was viewed as moderately large while the combination $n_1 = 30$ and $n_2 = 20$ was viewed as a relatively small sample size. The combination $n_1 = 10$ and $n_2 = 10$ was considered as a small sample.

Table 1 gives the observed coverages and the average lengths of the various confidence intervals for different proportions and different sample sizes. Table 2 presents the deviations from the definition intervals and zero width intervals. All values are given as percentages in Table 2. The results for the different interval methods are presented in the tables in sequence of computer intensiveness from the least to the most computer intensive method. The least computer intensive method is the usual Asymptotic interval method with or without continuity correction, and the most computationally intensive methods are the iterative methods of Mee, and of Miettinen and Nurminen.

3.2 Discussion of simulation study results

The ideal confidence interval method should result in a short interval with coverage close to the nominal coverage, should not deviate from the definition interval, and should have no occurrence of zero width intervals.

3.2.1 Case $p_1 = 0.6$ and $p_2 = 0.4$

All methods have relatively good coverage except the Asymptotic method without correction (As). The methods with continuity correction exceed the nominal coverage of the interval. When the sample size is small ($n_1 = n_2 = 10$) some of the methods deviate from the definition interval.

Table 1. Observed coverage and average length of nominal 95% confidence intervals

p_1, p_2	Sample size		Method													
	n_1, n_2		As	AsC	HA	Hal	JP	Sc	ScC	W	WC	B	K	Mee	MN	
0.6, 0.4	100, 90	Observed coverage (%)	94.7	96.2	95.7	95.1	95.1	94.9	96.1	95.1	96.6	95.1	95.7	95.1	95.1	
		Average length	.278	.299	.290	.275	.275	.272	.287	.275	.296	.276	.289	.275	.276	
	60, 50	Observed coverage (%)	94.5	96.5	95.9	95.1	95.1	94.8	96.2	95.1	96.8	95.2	95.8	95.1	95.1	
		Average length	.364	.401	.388	.358	.358	.353	.377	.358	.394	.360	.386	.358	.360	
	30, 20	Observed coverage (%)	93.8	96.6	95.8	94.7	94.7	94.4	96.6	94.7	97.6	94.9	95.8	94.7	94.7	
		Average length	.542	.625	.604	.523	.523	.506	.557	.521	.597	.530	.597	.523	.528	
	10, 10	Observed coverage (%)	92.5	97.3	96.2	95.6	95.6	95.6	97.4	95.6	98.9	96.4	96.2	95.9	95.9	
		Average length	.813	1.01	.957	.751	.755	.713	.813	.759	.930	.772	.955	.759	.777	
	0.95, 0.85	100, 90	Observed coverage (%)	94.4	97.0	96.1	94.7	95.1	95.4	97.2	94.7	97.2	99.9	96.8	95.2	95.2
			Average length	.169	.189	.181	.167	.170	.177	.191	.171	.193	.280	.186	.176	.177
60, 50		Observed coverage (%)	93.4	97.4	95.8	94.7	95.7	96.3	97.9	94.9	97.7	99.9	96.9	94.6	94.6	
		Average length	.223	.259	.245	.219	.226	.239	.265	.232	.271	.367	.254	.238	.239	
30, 20		Observed coverage (%)	92.5	95.5	94.5	94.5	94.5	96.7	98.9	93.7	98.5	99.8	95.9	92.2	92.2	
		Average length	.334	.417	.392	.322	.346	.379	.435	.378	.471	.541	.410	.377	.382	
10, 10		Observed coverage (%)	79.7	99.7	87.4	79.8	97.6	99.1	99.9	79.8	99.9	99.9	99.2	99.1	99.2	
		Average length	.451	.651	.576	.416	.519	.619	.737	.585	.903	.793	.656	.627	.646	
0.95, 0.15		100, 90	Observed coverage (%)	94.2	95.9	95.6	95.1	95.1	94.7	96.5	95.5	97.6	95.5	95.7	95.4	95.4
			Average length	.169	.189	.181	.170	.170	.172	.186	.177	.199	.171	.186	.169	.169
	60, 50	Observed coverage (%)	93.7	95.8	95.1	95.2	95.2	95.0	96.7	95.8	98.2	95.5	95.7	95.2	95.3	
		Average length	.222	.259	.245	.225	.225	.229	.253	.248	.288	.227	.254	.224	.225	
	30, 20	Observed coverage (%)	88.2	93.5	93.3	95.6	95.6	94.2	97.2	97.1	99.1	95.6	94.4	95.6	95.6	
		Average length	.333	.416	.391	.344	.344	.351	.401	.416	.503	.350	.410	.337	.341	
	10, 10	Observed coverage (%)	87.3	88.1	87.9	93.6	95.9	95.9	95.9	98.1	99.3	95.9	99.7	95.9	95.9	
		Average length	.451	.651	.575	.511	.514	.539	.639	.664	.850	.527	.656	.500	.516	

Table 1. Continued

p_1, p_2	Sample size		Method													
	n_1, n_2		As	AsC	HA	Hal	JP	Sc	ScC	W	WC	B	K	Mee	MN	
0.98, 0.98	100, 90	Observed coverage (%)	95.6	99.6	98.5	95.6	98.5	98.5	99.8	97.1	99.8	100	99.8	97.1	97.1	
		Average length	.076	.097	.088	.076	.086	.107	.124	.090	.117	.282	.110	.107	.107	
	60, 50	Observed coverage (%)	98.2	99.9	99.6	99.2	99.6	99.4	99.9	99.2	99.9	100	99.9	99.2	99.2	
		Average length	.095	.132	.116	.094	.119	.160	.189	.139	.187	.369	.156	.161	.162	
	30, 20	Observed coverage (%)	99.8	99.9	99.9	99.6	99.9	99.6	99.9	99.6	99.9	100	99.9	99.6	99.6	
		Average length	.119	.202	.171	.127	.205	.299	.363	.304	.420	.545	.269	.303	.308	
	10, 10	Observed coverage (%)	99.9	100	99.9	99.9	99.9	99.9	100	99.9	100	100	99.9	99.9	99.9	
		Average length	.133	.333	.240	.122	.380	.570	.699	.197	.919	.802	.479	.572	.592	
	0.98, 0.02	100, 90	Observed coverage (%)	88.9	97.7	89.3	93.8	93.8	92.6	96.1	94.2	98.5	96.1	97.7	96.1	96.1
			Average length	.076	.097	.088	.086	.086	.094	.108	.096	.116	.087	.109	.085	.085
		60, 50	Observed coverage (%)	88.6	88.8	88.7	97.0	97.0	93.1	94.2	97.8	99.1	97.0	99.9	95.7	95.7
			Average length	.095	.131	.115	.118	.118	.135	.158	.139	.171	.123	.156	.117	.118
		30, 20	Observed coverage (%)	63.6	99.9	63.7	92.2	94.9	93.4	93.8	95.7	99.2	98.0	99.9	96.1	96.1
			Average length	.119	.203	.172	.200	.203	.239	.288	.238	.296	.226	.269	.207	.209
		10, 10	Observed coverage (%)	32.8	100	100	94.0	94.0	94.0	96.6	95.5	32.5	94.0	100	94.0	94.0
			Average length	.131	.331	.238	.369	.369	.429	.526	.416	.519	.372	.478	.364	.378

As Asymptotic without correction; AsC Asymptotic with correction; B Belyaev; HA Asymptotic with Hauck and Anderson's correction; Hal Haldane interval; JP Jeffreys-Perks interval; K Kolmogorov; Mee Mee's interval; MN Miettinen and Nurminen interval; Sc Score interval without correction; ScC Score interval with correction; W Wallenstein's interval without correction; WC Wallenstein's interval with correction.

Table 2. Continued

p_1, p_2	Sample size n_1, n_2		Method											
			As	AsC	HA	Hal	JP	Sc	ScC	W	WC	B	K	Mee MN
0.98, 0.98	100, 90	Violation of definition interval (%)	0	0	0	0	0	0	0	0	0	0	0	0
		Zero width interval (%)	2.2	0	0	0	0	0	0	0	0	0	0	0
	60, 50	Violation of definition interval (%)	0	0	0	0	0	0	0	0	0	0	0	0
		Zero width interval (%)	10.7	0	0	0	0	0	0	0	0	0	0	0
	30, 20	Violation of definition interval (%)	0	0	0	0	0	0	0	0	0	0	0	0
		Zero width interval (%)	36.5	0	0	0	0	0	0	0	0	0	0	0
	10, 10	Violation of definition interval (%)	0	0	0	0	0	0	0	0	0	0	0	0
		Zero width interval (%)	66.6	0	0	66.6	0	0	0	66.6	0	0	0	0
0.98, 0.02	100, 90	Violation of definition interval (%)	44.9	81.7	66.9	0	2.2	0	0	25.3	26.3	2.24	85.4	0
		Zero width interval (%)	2.2	0	0	0	0	0	0	0	0	0	0	0
	60, 50	Violation of definition interval (%)	71.1	97.7	92.9	0	11.2	0	0	28.2	28.3	24.5	97.7	0
		Zero width interval (%)	11.2	0	0	0	0	0	0	0	0	0	0	0
	30, 20	Violation of definition interval (%)	61.9	99.9	99.7	0	36.3	0	0	20.2	20.3	73.5	99.9	0
		Zero width interval (%)	36.3	0	0	0	0	0	0	0	0	0	0	0
	10, 10	Violation of definition interval (%)	32.7	100	100	0	0	0	0	15.1	15.3	0	100	0
		Zero width interval (%)	67.2	0	0	0	0	0	0	0	0	0	0	0

As Asymptotic without correction; AsC Asymptotic with correction; B Belyaev; HA Asymptotic with Hauck and Anderson's correction; Hal Haldane interval; JP Jeffreys-Perks interval; K Kolmogorov; Mee Mee's interval; MN Miettinen and Nurminen interval; Sc Score interval without correction; ScC Score interval with correction; W Wallenstein's interval without correction; WC Wallenstein's interval with correction.

Methods recommended are the Haldane (Hal), Jeffreys-Perks (JP), Score interval without (Sc) and with correction (ScC), Wallenstein's interval without correction (W), Belyaev (B), Kolmogorov (K), Mee (Mee) and Miettinen and Nurminen (MN) intervals. Easy calculable methods are the Haldane interval, Jeffreys-Perks interval, Belyaev interval, Kolmogorov interval and the Score interval methods.

3.2.2 Case $p_1 = 0.95$ and $p_2 = 0.85$

Most of the methods have a good coverage when the sample size is at least moderately large. When the sample size is smaller the methods either over or under cover. Violation of the definition interval and zero width intervals also occur when the sample size is small. Continuity corrected methods tend to exceed the nominal coverage of the interval.

Methods recommended for large samples are the Jeffreys-Perks (JP), Score interval without (Sc) and with correction (ScC), Wallenstein's interval without correction (W), Kolmogorov interval (K), Mee (Mee) and Miettinen and Nurminen (MN) intervals.

The recommended methods for small samples are the Jeffreys-Perks, Kolmogorov interval and Score interval methods. The Mee and Miettinen and Nurminen intervals do not attain nominal coverage for sample size combination (30,20).

3.2.3 Case $p_1 = 0.95$ and $p_2 = 0.15$

Most methods have good coverage for large sample sizes. When the sample size gets smaller most of the methods either exceed or attain coverage less than the nominal coverage of the interval. Violation of the definition interval is common when the sample size is less than 100.

Methods recommended for large samples are the Asymptotic with Hauck and Anderson's correction (HA), Belyaev (B), Haldane (Hal), Jeffreys-Perks (JP), Mee (Mee) and Miettinen and Nurminen (MN) intervals.

Recommended methods for small samples are the Jeffreys-Perks, Haldane, Score interval without and with correction, Mee and Miettinen and Nurminen intervals.

3.2.4 Case $p_1 = 0.98$ and $p_2 = 0.98$

Most methods exceed the nominal coverage of the interval when the sample size gets smaller. Zero width intervals occur in some of the methods for the small sample size ($n_1 = 10 = n_2$).

Most methods exceed the nominal coverage, but taken length, deviations and zero width intervals into account the following methods could be used: methods recommended for large samples are the Asymptotic with Hauck and Anderson's correction (HA), Haldane (Hal), Jeffreys-Perks (JP), Score interval without (Sc) and with correction (ScC), Wallenstein's interval without correction (W), Mee (Mee) and Miettinen and Nurminen (MN) intervals; recommended methods for small samples are the Jeffreys-Perks, Score interval without and with correction, Mee and Miettinen and Nurminen intervals.

3.2.5 Case $p_1 = 0.98$ and $p_2 = 0.02$

Methods either exceed or do not attain the nominal coverage of the interval for large sample sizes. For smaller sample sizes this tendency is continued. Violation of the definition interval occurs in most of the methods for all sample size combinations.

It is difficult to recommend a method in this case as all methods either exceed or never attain the nominal coverage of the interval. Here the methods of Mee (Mee) and Miettinen and Nurminen (MN) may be used for large and

small sample sizes as both methods do not suffer from any deviations of definition interval or occurrence of zero width intervals. Both methods tend to exceed the nominal coverage of the interval for large samples and never attain the nominal coverage of the interval for the smallest sample size combination.

3.3 Recommendations

Beal (1987) noted that the Jeffreys-Perks interval is a definite improvement on the conventional interval. There are, however, some values of p_1 and p_2 for which this interval is too narrow, while the conventional approximate interval is not too narrow for these choices of p_1 and p_2 . According to Newcombe (1998), the Jeffreys-Perks and Haldane intervals are prone to certain novel anomalies such as overt overshoot, which occurs when the upper confidence limit is larger than 1 or the lower confidence limit is less than -1. He recommends the Score interval methods with and without correction.

In his more limited comparison of methods Beal (1987) came to the conclusion, that if one wants to implement a more sophisticated and complicated interval using a computer program, the Mee and Miettinen and Nurminen intervals are good choices. Beal recommended the use of the Jeffreys-Perks interval when one wants to compute a simple interval.

The Score interval methods have coverage close to 95% or higher and also relatively short lengths for all sample sizes. The Score interval with correction has a shorter length than the Score interval without correction, but both methods have no violations of the definition interval or intervals of zero width. The Score interval methods thus seem good methods to use in most practical situations.

We recommend the interval methods of Miettinen and Nurminen and Mee for use across all cases including small sample size. They are, however, computer intensive because of their iterative nature. The best of the more

easily calculable methods seem to be the Jeffreys-Perks and Score interval methods.

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